

		DEPARTMENT OF INFORMATION TECHNOLOGY	
		FACULTY OF COMPUTING	
MODULE OUTLINE			
Module Name		Mathematics for Computing	
Module Code		IT1130	Version No. 2024-1
Year/Level		1	Semester 1
Credit Points		4 Credits	
Pre-requisites		None	
Co-requisites		None	
Methods of Delivery	Lectures		2 Hours/Week
	Tutorials		1 Hours/Week
	Laboratories		2 Hours/Week
Courseweb Site		http://courseweb.sliit.lk	
Date of Original Approval		June, 2024	
Date of Last Approval		June, 2024	
Date of Next Review		October, 2028	
MODULE DESCRIPTION			
Aim	The aim of this module is to provide the students with an understanding of engineering mathematics required to solve engineering and IT problems. The students will learn trigonometry, calculus, solving equations, applying numerical methods in solving equations, applying complex numbers and matrices/determinants in solving real-world problems.		
Learning Outcomes	At the end of the module, the student will be able to:		
	LO1:	Apply differentiation integration and partial factors to solve scientific problems	
	LO2:	Apply complex numbers and trigonometry to solve real-world problems	
	LO3:	Analyze the solvability of equations using matrices and determinants	
	LO4:	Solve a set of linear equations using matrices	
	LO5:	Apply concepts in number systems computer arithmetic Logic and Boolean algebra	
Assessment Criteria	Assessment Description:		
	During the semester, there will be practical test, Lab submissions and a mid-term exam. The final examination will be a comprehensive exam based on the lecture materials covered during the semester		
	Assessment Activities:		
	Mid Exam	20%	LO1, LO2, LO5
	Practical Test	10%	LO3 - LO4
	Lab Submissions	10%	LO1 - LO5
	Final Examination	60%	LO1 - LO5
	Total	100%	
	Contact Hours		
	Lectures	28 Hours	
	Tutorials	14 Hours	
	Laboratories	28 Hours	
	Time Allocated for Assessments		
	Mid Exam	1 Hours	
	Practical Test	1 Hours	
	Final Examination	2 Hours	

	Reading and Independent Study	126 Hours
	Total	200 Hours
Module Pass Requirements	To pass this module, students need to obtain a 50% in “Continuous Assessments” and an overall mark that would qualify for a “C” grade or above	
Learning Resources	Recommended Texts 1. D. Jordan, P. Smith, Mathematical Techniques: An Introduction for the Engineering, Physical, and Mathematical Sciences, 4th Edition, Oxford University Press, 2008 2. K. A. Stroud and D. J. Booth, Engineering Mathematics, 8th Edition, Bloomsbury Academic, 2020	

CONTENTS OF THE MODULE		
1.	Logic Control - Number Systems (3 hours)	LO5
	a. Introduction to number bases, number base conversions and conversion of fractions	
2.	Logic Control - Boolean Algebra/Computer Arithmetic (3 hours)	LO5
	a. Introduction to Boolean Algebra b. Simplifying Boolean Expressions using Boolean Identities and deriving logical expressions	
3.	Logic Control - Logic Gates (3 hours)	LO5
	a. Introduction to logic gates b. Logic circuits design process	
4.	Fundamentals of Trigonometry (6 hours)	LO2
	a. Introduction to Partial Fractions b. Introduction to trigonometry	
5.	Introduction to Calculus (6 hours)	LO1
	a. Differentiation b. Integration	
6.	Complex number systems (10 hours)	LO2
	a. Different representations, addition, subtraction, division, multiplication, power of complex numbers and Argand Diagrams b. Complex conjugate and De Moivre’s Theorem and its applications c. <u>Write test cases and test the program</u>	
7.	Matrices and Determinants (15 hours)	LO3
	a. Vector Algebra, dot product, matrix operations, special matrices and their properties b. Inverse of a matrix, determinants, line integrals c. Surface integrals, eigenvalues, eigenvectors, diagonalization	
8.	Solving Linear Equations (15 hours)	LO4
	a. Echolen matrix method, inverse matrix approach b. Cramer’s rule	
9.	Numerical Methods (9 hours)	LO1,LO2
	a. One point iteration, Newton Raphson method b. Bisection method, Runga Kutta Method	

LEARNING OUTCOMES TO PROGRAM OUTCOMES MAPPING													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PIO	P11	P12	
LO1	X												
LO2	X												
LO3	X												
LO4	X												
LO5	X												
Module	X												
PO1	Apply knowledge of mathematics, statistics, basic sciences, and computer science to solve complex problems related to the computing domain.												
PO2	Identify, formulate, and analyze complex problems related to the computing domain by using established approaches in computing while recognizing their limitations, reviewing technical literature, and interpreting the results of experiments designed using appropriate research methods.												
PO3	Design systems, processes, or services to meet desired needs within realistic constraints.												
PO4	Apply reasoning informed by contextual knowledge in the area of information technology to assess the impacts of IT solutions in a global, economic, environmental, societal, legal, cultural, and health and safety context.												
PO5	Design, implement and deploy IT solutions using industry best practices including continuous improvements, quality assurance and mitigating security risks.												
PO6	Create, select, and apply appropriate computing tools, techniques and resources necessary to solve IT related real world problems												
PO7	Apply broad knowledge of sustainable development concepts and practices required for developing a solution in the information technology domain.												
PO8	Apply ethical principles and commit to professional ethics, responsibilities, and norms in the field of Computing												
PO9	Function effectively within a multidisciplinary and multicultural team which promotes equality, diversity, and inclusion.												
PO10	Communicate on computing activities with a wide variety of audiences by means of appropriate designs, plans, reports, presentations, and instructions.												
PO11	Apply broad knowledge of management principles, commercial context, change management, relevant legal matters including intellectual property rights and risk mitigation to manage projects in a multidisciplinary context.												
PO12	Plan and record self-learning and continuous professional development as the foundation for lifelong learning.												
GENERAL INFORMATION													
Any type of plagiarism is not allowed.													
Plagiarism: Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the Institute as whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Plagiarism occurs when the work of another person, or persons, is used and presented as one's own.													
END OF MODULE OUTLINE													